Fixed-Parameter Additive Gaps for Deterministic Communication Complexity

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Abstract

How can a one-bit gap in deterministic communication complexity be amplified without making the underlying matrix too large? We give a finite construction that achieves an arbitrary fixed additive gap while controlling both matrix dimensions. Relative to a stated source theorem and a balanced-family theorem, we prove the following for every fixed positive integer k. Every valid ordered 3-CNF formula Φ whose number n of variables exceeds a cutoff depending only on k and the fixed inputs yields a nonempty family of Boolean matrices with integer budgets. Satisfiability implies that some matrix has communication complexity at most its budget, whereas unsatisfiability implies that every matrix has complexity exactly its budget plus k. The family is defined without knowing which case holds. Every matrix has at most $2^{\lfloor n/(2k) \rfloor}$ rows and columns, and hence at most $2^{\lfloor n/k \rfloor}$ entries.

The main difficulty is preserving a lower bound under the restrictions needed for repeated amplification. We duplicate selected parts of a one-bit hardness construction so that its lower bounds survive deletions, then use an interlace and transposition to bring two retention thresholds into agreement. A general amplification lemma restores the property needed for the next level. Each level increases the gap by exactly one, and exact row and column recurrences control the size after k levels.

1 Introduction

In deterministic communication complexity, Alice receives a row of a Boolean matrix M, Bob receives a column, and they exchange bits to determine the corresponding entry. The communication complexity $D(M)$ is the minimum worst-case number of bits required by such a protocol [3, 4]. Throughout, row and column sets are finite and nonempty.

We study how to amplify a gap between two possible communication costs while controlling the dimensions of the matrix. A construction that separates the two cases by one bit need not remain useful when repeated: an intermediate restriction may lose the lower bound that the next application requires. The central task is therefore to increase the gap and, at the same time, preserve sufficiently strong lower bounds on large restrictions.

Starting from the companion hardness construction [2], we obtain an arbitrary fixed additive gap k. For sufficiently large valid input formulas, the construction produces a nonempty family of matrices with associated budgets. A satisfiable formula gives a matrix whose complexity is at most its budget; an unsatisfiable formula makes every matrix's complexity exactly k larger. The matrices have at most $2^{\lfloor n/k \rfloor}$ entries for sufficiently large formulas with n variables. Both the amplification and this size bound concern the same family.

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1.1 Our result

The input to our construction is an *ordered* 3-CNF formula Φ: the clause order, the three positions within each clause, and repeated literals are part of the input. Let $n(\Phi)$ be its declared number of variables. For $k \geq 1$, put

$$e(n,k) = \left\lfloor \frac{n}{2k} \right\rfloor, \qquad S(n,k) = 2^{e(n,k)}, \qquad E(n,k) = S(n,k)^2. \tag{1}$$

For a nonempty finite matrix define its *naming width* by

$$w(M) = \max\left\{1, \lceil \log_2 |\mathsf{Rows}(M)| \rceil, \lceil \log_2 |\mathsf{Cols}(M)| \rceil\right\}. \tag{2}$$

This measures the larger number of bits needed to name a row or a column, with a minimum value of one. The natural-number ceiling logarithms also cover one-element row and column sets.

We call a construction *promise-independent* when its matrices and budgets are determined by the input formula without testing satisfiability, using a satisfying assignment, or assuming unsatisfiability. The two cases below are properties of one construction, not separate choices of matrices.

Theorem 1.1 (Finite fixed-parameter gap and same-matrix size)**.** *Fix a composed finite source theorem and a balanced-family theorem satisfying the conditions in Section 2.2. There is one promise-independent construction with the following property. For every $k \geq 1$ there is an integer $U_k \geq k$, chosen before the formula, such that every valid ordered 3-CNF formula Φ with $n(\Phi) \geq U_k$ produces a positive integer $t(\Phi)$ and a family*

$$(M_{\Phi,k,j}, Y_{\Phi,k,j}), \qquad j \in [t(\Phi)].$$

Every matrix in the family has nonempty row and column sets, and the entire family is fixed before satisfiability is used. Moreover,

$$\Phi \text{ satisfiable} \Longrightarrow \exists j \in [t(\Phi)] \quad D(M_{\Phi,k,j}) \leq Y_{\Phi,k,j}, \tag{3}$$

$$\Phi \text{ unsatisfiable} \Longrightarrow \forall j \in [t(\Phi)] \quad D(M_{\Phi,k,j}) = Y_{\Phi,k,j} + k, \tag{4}$$

$$\forall j \in [t(\Phi)] \qquad w(M_{\Phi,k,j}) \leq e(n(\Phi),k). \tag{5}$$

For those same matrices,

$$|\mathsf{Rows}(M_{\Phi,k,j})| \leq S(n(\Phi),k), \qquad |\mathsf{Cols}(M_{\Phi,k,j})| \leq S(n(\Phi),k), \tag{6}$$

$$|\mathsf{Rows}(M_{\Phi,k,j})|\,|\mathsf{Cols}(M_{\Phi,k,j})| \leq E(n(\Phi),k) \leq 2^{\lfloor n(\Phi)/k \rfloor}. \tag{7}$$

The cutoff U_k is uniform over the formula and every graph supplied by the source theorem. The size bounds apply directly to the matrices in the gap statement; no padding is used. Our result concerns this finite construction and its dimensions, without a running-time claim.

1.2 The obstacle to repeated amplification

The fourth stage of the companion construction creates a one-bit gap, but cannot immediately be used again. To repeat the construction, we need lower bounds that survive specified deletions of rows and columns. In particular, two lower bounds must hold at a common, small row-retention threshold. The unmodified matrix does not preserve its full communication cost under such restrictions.

Two obstructions explain the difficulty. An off-diagonal column restriction retains at least $7/8$ of the columns while making all entries that encode the source graph neutral. Independently, if an optimal protocol begins with Alice, one child keeps at least half the rows and has remaining cost at most one bit below the original cost. Thus a lower bound at the full original cost cannot simply be demanded for every sufficiently dense restriction.

We overcome this difficulty in two steps. First, we duplicate selected rows and columns. The resulting matrix N retains its full cost when only a small fraction of rows and columns is deleted, and a specified reference restriction satisfies three further lower bounds, beginning one bit below that cost. Second, an interlace followed by transposition produces a matrix W for which the two kinds of estimates hold at the same row threshold. We call this step *scale alignment.* A general lemma then restores the restriction property needed for another application of the construction. The complete level increases the gap by exactly one bit.

The complete proof has the following form:

$$\begin{array}{c} \text{robust full-cube seed} \\ \downarrow \\ \text{Stages 1–3 and raw Stage 4} \\ \downarrow \\ \text{protected Stage 4} \\ \downarrow \\ \text{scale alignment} \\ \downarrow \\ \text{generic one-bit bridge} \\ \downarrow \\ k\text{-level recursion and exact carrier induction.} \end{array}$$

Section 2 defines the restriction properties and states the two outside results. Section 3 gives one level of the matrix construction, and Section 4 proves the amplification step. Section 5 controls the gap and the dimensions through the recursion. Section 6 combines these ingredients to prove the main theorem. The appendices give the detailed lower-bound arguments and parameter estimates.

2 Preliminaries and auxiliary results

For $m \in \mathbb{N}$, write $[m] = \{0, \ldots, m-1\}$. The restriction $M|_{A\times B}$ has row set A and column set B; these sets are always nonempty when its communication complexity is used. Transposition and bijective relabelling preserve communication complexity; restriction cannot increase it; and duplicating equal rows or columns preserves it if one representative of every original input remains.

We use one elementary lower bound repeatedly. In a reduced depth-d protocol for a nonconstant Boolean matrix, at most 2^{d-1} leaves have any fixed output bit. The entries reaching each such leaf form a rectangle, so if the real rank of the indicator matrix of one output exceeds 2^{d-1}, then the communication complexity is at least $d + 1$.

2.1 Interlaces and lower bounds under restriction

For $p \geq 1$ and $M : X \times Y \to \{0, 1\}$, define

$$\mathrm{Int}_p(M)((i, x), f) = M(x, f(i)), \quad (i, x) \in [p] \times X, \quad f : [p] \to Y. \tag{8}$$

Its two carrier sizes are

$$|\mathsf{Rows}(\mathrm{Int}_p(M))| = p|X|, \qquad |\mathsf{Cols}(\mathrm{Int}_p(M))| = |Y|^p. \tag{9}$$

If p is a positive power of two, the parties can first communicate the coordinate i and then run a protocol for M, giving

$$D(\mathrm{Int}_p(M)) \le D(M) + \log_2 p. \tag{10}$$

We need lower bounds that remain valid after deleting rows and columns. Let $\mathcal{D}_M(p; x, z)$ be the minimum communication complexity among restrictions of $\mathrm{Int}_p(M)$ that retain at least $\lceil x|X| \rceil$ rows in *each* outer copy and at least $\lceil z|Y|^p \rceil$ columns. The row threshold is per copy, not merely a total threshold. Define

$$\Lambda_M(p; x, z) = \min\{\mathcal{D}_M(p; x, z), 1 + \mathcal{D}_M(p; x, z/2), 2 + \mathcal{D}_M(p; x, z/4)\} \tag{11}$$

and

$$\mathsf{Rob}(M; b, z) = \Lambda_M(1; 2^{-b}, z). \tag{12}$$

Thus $C \le \mathsf{Rob}(M; b, 3/5)$ simultaneously says that the three retained-column densities $3/5, 3/10, 3/20$ have lower bounds $C, C-1, C-2$. Bijections of both carriers preserve $\mathcal{D}$, Λ, and Rob.

2.2 The composed source theorem

An ordered multigraph G has ordered vertex and edge-occurrence carriers, with endpoint maps $\partial_0, \partial_1 : [e(G)] \to [v(G)]$. Parallel edge occurrences remain distinct. For a colouring $c : [v(G)] \to [3]$, let $\mathrm{mono}(G, c)$ count monochromatic edge occurrences.

The source theorem supplies fixed integers $c_{\mathrm{src}}, \Delta, n_{\mathrm{src}} \ge 1$, a fixed rational $0 < \gamma \le 1$, and, for every ordered formula Φ, one finite ordered list $\mathcal{S}(\Phi)$ of graphs. For every valid Φ with $n = n(\Phi) \ge n_{\mathrm{src}}$:

(S1) the list has positive length $t(\Phi)$;

(S2) every listed graph is nonempty, loop-free, and has every degree between 1 and Δ;

(S3) every listed graph has at least n edge occurrences and satisfies

$$v(G) + e(G) \le n\big(\log_2(n+2)\big)^{c_{\mathrm{src}}};$$

(S4) if Φ is satisfiable, some listed graph is properly three-colourable;

(S5) if Φ is unsatisfiable, then for every listed graph and every three-colouring c, $\gamma e(G) \le \mathrm{mono}(G, c)$.

The same ordered list occurs in all five clauses. These properties combine the source transformations needed below into one finite statement, which we take as an auxiliary input. Its intended source is the companion construction [2]. Expanders may be used to establish this source theorem, but no expander is an additional input to the matrix construction.

It is convenient to replace the real logarithm by an integer logarithm. For $n \ge 2$, put $L = \lfloor \log_2(n+2) \rfloor \ge 2$. Then $\log_2(n+2) < L + 1 \le L^2$. Replacing c_{src} by $2c_{\mathrm{src}}$ and n_{src} by $\max\{n_{\mathrm{src}}, 2\}$ therefore yields the integer-logarithm bound $v(G) + e(G) \le nL^{2c_{\mathrm{src}}}$, with the original exponent on the right. The ordered graph lists, γ, Δ, and both promise claims are unchanged. We make this replacement before selecting parameters and reuse the names $c_{\mathrm{src}}, n_{\mathrm{src}}$ for the normalized constants.

2.3 The balanced-family theorem

For $m > 0$ and $1 \le t \le q$, an indexed family is $\mathcal{F} = (f_\ell : [q] \to [m])_{\ell \in [N]}$. Given a set of coordinates $J \subseteq [q]$ and a pattern $a : J \to [m]$, write

$$C_{\mathcal{F}}(J, a) = \left|\{\ell \in [N] : f_\ell|_J = a\}\right|.$$

The balanced-family theorem supplies one such family for every legal (q, t, m), with $N > 0$, all entries in $[m]$, and

$$N < \left(2 \max\{q, 1000mt^2, 2\}\right)^t, \tag{13}$$

$$(100t - 1)N \le 100t\, m^{|J|} C_{\mathcal{F}}(J, a) \le (100t + 1)N \quad (|J| \le t). \tag{14}$$

These families are related to small almost-independent sample spaces [1]. We use the following explicit polynomial construction because it also gives the product estimates required in the lower-bound proof.

We will sample independently from several copies of the same family. The scale-alignment argument requires three estimates for these products. For a family $\mathcal{C} = (c_\ell : [Q] \to \mathcal{Y})_{\ell \in [L]}$, let $\mathsf{Bal}_t(\mathcal{C}; \varepsilon)$ mean that $L > 0$ and, for every $J \subseteq [Q]$ with $r = |J| \le t$ and every total pattern $a : [Q] \to \mathcal{Y}$,

$$\left| \frac{|\{\ell : c_\ell(\alpha) = a(\alpha) \text{ for all } \alpha \in J\}|}{L} - \frac{1}{|\mathcal{Y}|^r} \right| \le \frac{\varepsilon}{|\mathcal{Y}|^r}. \tag{15}$$

For a finite set D indexing the copies, the product family $\mathcal{C}^{\times D}$ is indexed by maps $\lambda : D \to [L]$, has coordinates $(d, \alpha) \in D \times [Q]$, and has entry

$$\mathcal{C}^{\times D}_\lambda(d, \alpha) = c_{\lambda(d)}(\alpha). \tag{16}$$

At one level of the construction, let $\mathcal{C}$ be the Stage–2 family, with parameters q_2, t_2, N_2 and alphabet X_1. Put

$$\varepsilon_2 = \frac{1}{100t_2}, \qquad \varepsilon_{\text{prod}} = \max\left\{(1 + \varepsilon_2)^{t_2} - 1, 1 - (1 - \varepsilon_2)^{t_2}\right\}. \tag{17}$$

We require the following three product estimates:

(B1) for every $e \ge 1$ and every $J' \subseteq [2^e]$ with $|J'| = 2^{e-1}$, the product over the donor subtype $D_{e,J'} = J'$ satisfies $\mathsf{Bal}_{t_2}(\mathcal{C}^{\times D_{e,J'}}; \varepsilon_2)$;

(B2) the two-donor product satisfies $\mathsf{Bal}_{t_2}(\mathcal{C}^{\times [2]}; \varepsilon_{\text{prod}})$;

(B3) for the canonical exceptional exponent $e_{\text{exc}} \ge 2$, every $J' \subseteq [2^{e_{\text{exc}}}]$ with $|J'| = 2^{e_{\text{exc}}-1}$ again satisfies $\mathsf{Bal}_{t_2}(\mathcal{C}^{\times D_{e_{\text{exc}},J'}}; \varepsilon_2)$.

All three are counting properties of the chosen family, not communication lower bounds. The source list and the balanced families are selected independently of whether the formula is satisfiable.

A polynomial construction. Let $X = \max\{q, 1000mt^2, 2\}$ and choose the least prime p with $X < p < 2X$, which exists by Bertrand's postulate. Take all p^t polynomials of degree less than t over $\mathbb{F}_p$, evaluate them at q distinct residues, and map each value's representative in $[p]$ to its residue modulo m. This gives exactly the required size bound. Each alphabet bucket has size $b_a \in \{\lfloor p/m \rfloor, \lceil p/m \rceil\}$, so $|mb_a/p - 1| \le \delta = m/p < 1/(1000t^2)$.

Polynomial interpolation makes any $s \le t$ distinct field evaluations independent and uniform. In an arbitrary donor product, a pattern on $r \le t$ distinct donor-coordinate pairs therefore has normalized count
$$\frac{m^r C(J,a)}{(p^t)^{|D|}} = \prod_{(d,j)\in J} \frac{m b_{a(d,j)}}{p} \in [(1-\delta)^r, (1+\delta)^r].$$
Only r, not the number of donors, controls the error. Indeed, $(1-\delta)^r \ge 1 - r\delta$, while the binomial expansion gives $(1+\delta)^r \le \sum_{i=0}^{r} (r\delta)^i \le (1-r\delta)^{-1}$. Both errors are at most $1/(100t)$, since $r\delta \le 1/(1000t)$ and $r\delta/(1-r\delta) \le 1/(1000t-1) \le 1/(100t)$. Empty patterns and empty donor sets have normalized count 1. This proves (B1) and (B3) for the same chosen family; (B2) follows because $\varepsilon_{\mathrm{prod}} \ge \varepsilon_2$ by Bernoulli's inequality. Relabelling the alphabet and coordinate carriers changes none of these counts. For parameter requests outside the stated range, define the output to be a singleton all-zero family; this case is not used by the theorem. The argument above depends on choosing this polynomial family. The coarser estimate (14) alone would not preserve the same error under products of an arbitrary family.

3 One level of the construction

Fix one graph from the source list, write $S = (G, \gamma, \Delta)$, and let $R : [m_R] \times [n_R] \to \{0,1\}$ be the input matrix. One level uses six natural parameters
$$(a, s, q_2, t_2, q_{\mathrm{al}}, Q_{\mathrm{br}}). \tag{18}$$
The first four control the basic matrix construction, q_{al} is the interlace width used for scale alignment, and Q_{br} is the interlace width used by the final amplification lemma. The three widths q_2, q_{al}, and Q_{br} are chosen separately.

The structural assumptions are: $v(G), m_R, n_R > 0$; $a \ge 2$ and $s \ge 4$;
$$q_2 = 2^{\lceil \log_2(12se(G)) \rceil}, \qquad 1 \le t_2, \qquad 4t_2 \le q_2; \tag{19}$$
every active source item has rank below s at its associated coordinate; and $q_{\mathrm{al}}, Q_{\mathrm{br}} > 0$. These conditions depend only on the input data. To define the construction for all parameters, use a fixed all-false matrix on the prescribed row and column sets when a condition fails. Beyond the cutoff established in Section 5, all conditions hold at every level.

3.1 Stages 1–3

Put $q_1 = 2^a - 2$ and $L_1 = q_1 + s + 1$. The first stage is
$$X_1 = [q_1] \times [m_R], \qquad Y_1 = ([L_1] \to [n_R]),$$
$$M_1((i,x),\Gamma) = R(x, \Gamma(i)). \tag{20}$$
Identify $[q_1 m_R]$ with X_1 by quotient and remainder. From the balanced family $\mathcal{F}(q_2, t_2, q_1 m_R)$, of size N_2, obtain entries $c_{\ell,\alpha} \in X_1$. Then
$$X_2 = [q_2] \times Y_1, \qquad Y_2 = [N_2], \qquad M_2((\alpha,\Gamma),\ell) = M_1(c_{\ell,\alpha},\Gamma), \tag{21}$$
$$X_3 = [s] \times Y_2, \qquad Y_3 = ([s] \to X_2), \qquad M_3((p,\ell),d) = M_2(d(p),\ell). \tag{22}$$
Thus the third stage is the s-copy interlace of M_2^{T}.

3.2 Encoding the source graph in Stage 4

Let $V = [v(G)]$ and $E = [e(G)]$. The source rows use the vertices of G and $s-3$ additional items, called *blockers*. Coordinates distinguish an edge occurrence from each of its two incidences:

$$\mathcal{I} = V \sqcup [s-3], \qquad \mathcal{Z} = E \sqcup (E \times [2]).$$

Take $4s$ labelled copies of this incidence structure, called *slabs*. Order the local coordinates by

$$\operatorname{rank}_{\mathcal{Z}}(e) = e, \qquad \operatorname{rank}_{\mathcal{Z}}(e, b) = e(G) + 2e + b.$$

A vertex item accepts the edge coordinates on which it is an endpoint and the incidence coordinate that names it; a blocker accepts every local coordinate. Slab labels must agree. The local coordinate z in slab h is encoded by $h(3e(G)) + \operatorname{rank}_{\mathcal{Z}}(z) \in [q_2]$. The resulting local relation gives a guarded source matrix

$$G_{\mathrm{src}} : ([4s] \times \mathcal{I}) \times [q_2] \to \{0, 1\}. \tag{23}$$

At each coordinate, order the active source rows and assign them ranks starting at zero. The structural assumptions place these ranks in $[s]$.

The raw fourth-stage carriers are

$$\begin{aligned} X_B &= ([s] \times Y_2) \sqcup (([4s] \times \mathcal{I}) \times [m_R]), \\ Y_B &= ([s+1] \to [n_R]) \times ([s] \to X_2). \end{aligned} \tag{24}$$

Write a raw column as (β, d), and let α be the Stage–2 coordinate advertised at its first selector position. Template rows copy M_2:

$$B((p, \ell), (\beta, d)) = M_2(d(p), \ell). \tag{25}$$

For a source item (h, i) and seed row x, if all $d(p)$ advertise the same α and $G_{\mathrm{src}}((h, i), \alpha) = 1$, the source row outputs $R(x, \beta(r_\alpha(h, i)))$, where $r_\alpha(h, i) \in [s]$ is its active rank. Otherwise it outputs the neutral value $R(x, \beta(s))$. This rule defines every entry without using a colouring or assuming that no colouring exists.

3.3 Duplication and the two interlaces

Duplicating rows and columns does not change communication complexity, but it changes the effect of deleting a given fraction of them. We duplicate four groups of rows and three groups of columns of B with different multiplicities. Let

$$\begin{aligned} &T = [s] \times Y_2, && V_S = ([4s] \times V) \times [m_R], && B_S = ([4s] \times [s-3]) \times [m_R], \\ &S_C = ([s] \to X_2), && D_C = [q_2] \times Y_1. \end{aligned}$$

Each group is mapped to rows or columns of B as follows. The three additional row groups use their displayed summands of X_B; a selector $d \in S_C$ is paired with the constant seed tuple at column 0; and $(\alpha, \Gamma) \in D_C$ is sent to the column whose seed tuple is the tail of Γ and whose selector is constantly (α, Γ). Write

$$\begin{aligned} &M_0 = |Y_B|, \quad S_0 = |S_C|, \quad D_0' = |D_C|, \\ &X_0 = |X_B|, \quad T_0 = |T|, \quad V_0 = |V_S|, \quad B_0 = |B_S|. \end{aligned}$$

The column multiplicities are

$$w_{\text{base}} = 182 S_0 D_0', \qquad w_{\text{sel}} = M_0 D_0', \qquad w_{\text{diag}} = M_0 S_0, \tag{26}$$

and the row multiplicities are

$$\begin{aligned} u_{\text{base}} &= (2^{22} - 4) T_0 V_0 B_0, \qquad u_{\text{temp}} = 2 X_0 V_0 B_0, \\ u_{\text{vert}} &= X_0 T_0 B_0, \qquad u_{\text{block}} = X_0 T_0 V_0. \end{aligned} \tag{27}$$

The duplicated matrix A has the following disjoint row and column sets, with the second coordinate recording the copy label:

$$\begin{aligned} X_A &= (X_B \times [u_{\text{base}}]) \sqcup (T \times [u_{\text{temp}}]) \sqcup (V_S \times [u_{\text{vert}}]) \sqcup (B_S \times [u_{\text{block}}]), \\ Y_A &= (Y_B \times [w_{\text{base}}]) \sqcup (S_C \times [w_{\text{sel}}]) \sqcup (D_C \times [w_{\text{diag}}]). \end{aligned} \tag{28}$$

Forgetting labels and using the fixed inclusions into X_B, Y_B defines surjections ρ, κ, and

$$A(r, c) = B(\rho(r), \kappa(c)). \tag{29}$$

Copies of the same original row or column are equal. Since every original row and column is represented, these duplications preserve communication complexity. We call A the *protected matrix*, because the chosen multiplicities will make the required lower bounds survive deletions.

Now set

$$N = A^{\mathsf{T}}, \qquad U = \mathrm{Int}_{q_{\text{al}}}(N), \qquad W = U^{\mathsf{T}}, \qquad R^{+} = \mathrm{Int}_{Q_{\text{br}}}(W)^{\mathsf{T}}. \tag{30}$$

The reference matrix N_{ref} is the restriction of N whose rows are *all* base-labelled copies $Y_B \times [w_{\text{base}}]$ and whose columns are all of X_A. Hence

$$|\mathsf{Cols}(N_{\text{ref}})| = |\mathsf{Cols}(N)|. \tag{31}$$

Finally, relabel the rows and columns of R^{+} by the canonical bijections to zero-based integer intervals. This defines the successor used in the recursion.

3.4 Lower bounds for the protected matrix

Suppose that the graph satisfies the unsatisfiable-case conclusion (S5). Let $D_0 = D(R)$, and put

$$\begin{aligned} C &= D_0 + a + \log_2 q_2 + \log_2 s + 1, \\ C_W &= C + \log_2 q_{\text{al}}. \end{aligned} \tag{32}$$

The parameter choices in Section C fix positive deletion tolerances η_R, η_C, a reference depth b_T, and the numerical inequalities used below. Assume that the input matrix satisfies

$$D_0 \geq 3, \qquad D(R) = D_0, \qquad D_0 \leq \mathsf{Rob}(R; b_0, 3/5), \tag{33}$$

together with the graph properties (S2)–(S3) and the far-from-three-colourable condition (S5).

Theorem 3.1 (Lower bounds after protected duplication)**.** *Under the structural conditions, the seed assumptions, and the numerical conditions above, N and N_{ref} above satisfy*

$$C \leq \mathcal{D}_N(1; 1 - \eta_R, 1 - \eta_C), \tag{34}$$

$$D(N) = C, \tag{35}$$

$$C - 1 - r \leq \mathcal{D}_{N_{\text{ref}}}(1; 2^{-b_T}, (3/5)/2^r) \quad (r = 0, 1, 2). \tag{36}$$

The proof is given in Section A. Every row and column used in its lower-bound argument lies in the retained restriction of the matrix N defined above. Notice the different roles of the conclusions: (34) preserves the full cost under small deletions, whereas the three lower-density estimates (36) concern only N_{ref} and begin one bit lower.

For exactness, the dense statement applied to the full matrix gives $D(N) \geq C$. Conversely, the explicit raw protocol sends the a first-stage selector bits, the $\log_2 q_2$ second-stage coordinate bits, the $\log_2 s$ third-stage selector bits, and one bit distinguishing template and source rows before running an optimal protocol for R. Duplication and transpose preserve cost, so $D(N) \leq C$.

4 Scale alignment and the one-bit bridge

The protected matrix has two useful lower bounds, but their row-retention thresholds differ. The first interlace and transposition bring them to a common threshold on W, as defined in (30).

Choose

$$\delta_U = \frac{\eta_R}{16 q_{\mathrm{al}}}, \qquad Q_{\mathrm{br}} = \mathsf{Qwidth}(b_+, \delta_U), \qquad \theta = \frac{1}{160 Q_{\mathrm{br}}}, \tag{37}$$

where $\mathsf{Qwidth}(b, \delta)$ is the least positive power of two Q such that $80b \leq Q\delta$. The parameter choices ensure $\delta_U \leq 1/64$ and the remaining projection and counting inequalities listed in Appendix C.

Theorem 4.1 (Lower bounds at a common row threshold)**.** *Under the hypotheses of Theorem 3.1, the matrix $W = \mathrm{Int}_{q_{\mathrm{al}}}(N)^{\mathsf{T}}$ satisfies*

$$D(W) = C_W, \tag{38}$$

$$C_W \leq \mathcal{D}_W(1; \theta, 1 - \delta_U), \tag{39}$$

$$C_W - 1 \leq \Lambda_W(1; \theta, 3/5). \tag{40}$$

If $D(N) \leq Y_N$, then

$$D(W) \leq Y_N + \log_2 q_{\mathrm{al}}. \tag{41}$$

To prove (39), we work before transposition. Its equivalent statement is that, for every $R_U \subseteq \mathsf{Rows}(U)$ and $C_U \subseteq \mathsf{Cols}(U)$ satisfying

$$|R_U| \geq \lceil (1 - \delta_U)|\mathsf{Rows}(U)| \rceil, \qquad |C_U| \geq \lceil \theta |\mathsf{Cols}(U)| \rceil,$$

every protocol for $U|_{R_U \times C_U}$ has cost at least C_W. Transposition exchanges the row and column retention thresholds.

Here is the argument. A restriction of $U = \mathrm{Int}_{q_{\mathrm{al}}}(N)$ under consideration keeps all but a δ_U fraction of the rows and at least a θ fraction of the columns. Write R_i for the rows retained in coordinate i. Summing the row losses gives

$$\sum_{i < q_{\mathrm{al}}} (|\mathsf{Rows}(N)| - |R_i|) \leq \delta_U q_{\mathrm{al}} |\mathsf{Rows}(N)| = \frac{\eta_R}{16} |\mathsf{Rows}(N)|. \tag{42}$$

The same bound controls loss from the copies of the reference row set. Elementary product-projection estimates remove a bounded exceptional set of coordinates.

Suppose a protocol has cost at most $C_W - 1$. Follow its Alice questions until Bob first speaks or until $\log_2 q_{\mathrm{al}}$ Alice answers have been seen. Early leaves receive only a virtual route label; no query is added and the leaf remains unchanged. The resulting stopping routes are prefix-free, their row sets partition the retained rows, and the remaining subprotocol at a route of length d has cost at most $C_W - 1 - d$.

The proof keeps track of two different quantities: reference rows that witness a lower bound in a coordinate, and the total number of protected rows from that coordinate. At a node where Bob first speaks, either one of the three reference bounds applies or sufficiently many protected rows remain for the first product estimate (B1) to apply.

At the final Alice paths, a surviving reference-row witness determines a unique path, which we call its *owner*. A large fraction of the protected rows from the same coordinate must reach other paths. The two-copy product estimate (B2) shows that little of this fraction can reach another owner. Many rows must therefore reach the remaining, exceptional paths. Two applications of the pigeonhole principle, followed by (B3), produce a hard restriction inside one exceptional path. Its communication complexity exceeds the remaining cost of the protocol, a contradiction. Appendix A gives the counting argument, keeping reference witnesses and protected-row counts separate.

Minimizing over retained members proves (39). The full matrix gives $D(W) \geq C_W$, while the selector protocol and (35) give the reverse inequality. Applying the three reference rungs coordinatewise gives (40). The same selector protocol proves (41) from any upper bound on N.

4.1 A construction-independent bridge

Call the three statements

$$D(W) = C_W, \qquad C_W \leq \mathcal{D}_W(1;\theta,1-\delta), \qquad C_W - 1 \leq \Lambda_W(1;\theta,3/5) \tag{43}$$

the *common-threshold property.* This name only abbreviates the three displayed assertions.

Theorem 4.2 (Common-threshold bridge)**.** *Let W have nonempty finite carriers, let $C_W \geq 3$ and $b \geq 1$, and let $0 < \theta$ and $0 < \delta \leq 1/64$. Assume (43). If $Q = \mathsf{Qwidth}(b,\delta)$ and $Q\theta < 1/20$, then*

$$R^+ = \mathrm{Int}_Q(W)^{\mathsf{T}} \tag{44}$$

satisfies

$$D(R^+) = C_W + \log_2 Q, \tag{45}$$

$$C_W + \log_2 Q \leq \mathsf{Rob}(R^+; b, 3/5). \tag{46}$$

Moreover, $D(W) \leq Y_W$ implies

$$D(R^+) \leq Y_W + \log_2 Q. \tag{47}$$

The bridge does not depend on the source graph or on how W was constructed. Its proof, in Appendix B, uses an induction whose hypotheses control the logarithmic loss of column density under projections. The near-full one-copy estimate gives the base case; two- and three-copy consequences of the lower profile give the first increments. Thereafter, the first query of a protocol reduces the claim to a smaller one. A final count, including the ceiling terms, gives the requested depth b. The selector protocol supplies the matching upper bound and hence the equality.

For our parameter choices, $Q = Q_{\mathrm{br}}$ and $Q_{\mathrm{br}}\theta = 1/160 < 1/20$. Thus the bridge applies directly to R^+ from (30).

4.2 The complete one-level statement

Define the level overhead

$$H = a + \log_2 q_2 + \log_2 s + \log_2 q_{\mathrm{al}} + \log_2 Q_{\mathrm{br}}. \tag{48}$$

All three widths are positive powers of two in the canonical schedule.

Corollary 4.3 (One-level amplifier)**.** *The successor R^+ is fixed from the input matrix, source graph, parameters, and balanced-family entries before the source promise is used.*

(i) If the graph is properly three-colourable and $D(R) \leq Y$, then

$$D(R^+) \leq Y + H. \tag{49}$$

(ii) If the graph satisfies the far-from-three-colourable condition in (S5) and the seed and numerical conditions hold, then

$$D(R^+) = D_0 + H + 1, \tag{50}$$

$$D_0 + H + 1 \leq \mathsf{Rob}(R^+; b_+, 3/5). \tag{51}$$

(iii) If $m_N = |\mathsf{Rows}(N)|$ and $n_N = |\mathsf{Cols}(N)|$, the successor has

$$|\mathsf{Rows}(R^+)| = (q_{\mathrm{al}} m_N)^{Q_{\mathrm{br}}}, \tag{52}$$

$$|\mathsf{Cols}(R^+)| = Q_{\mathrm{br}} n_N^{q_{\mathrm{al}}}. \tag{53}$$

When the graph is three-colourable, the colouring removes the final template/source distinguishing bit from the raw upper protocol. The two interlace selector protocols then give (49). For graphs in the far-from-three-colourable case, the protected and scale-aligned lower bounds followed by Theorem 4.2 give exactly one additional bit. The carrier equations follow directly from (9) and canonical relabelling.

5 Iteration and matrix dimensions

Fix one formula Φ, one source input $S = (G, \gamma, \Delta)$, and

$$b = \lceil \log_2(n(\Phi) + 2) \rceil . \tag{54}$$

The graph G is chosen from the source list and remains fixed throughout the recursion.

5.1 A robust full-cube seed

Let $a_{\mathrm{seed}}(b)$ be the least $a \geq 3$ such that

$$64(b+1) \leq 2^a - 2, \tag{55}$$

and put $q_{\mathrm{seed}} = 2^{a_{\mathrm{seed}}(b)} - 2$. Define

$$F_b(x, i) = x(i), \qquad x : [q_{\mathrm{seed}}] \to \{0,1\}, \quad i \in [q_{\mathrm{seed}}]. \tag{56}$$

Its rows are all Boolean vectors and its columns are coordinates. If $y_0 = a_{\mathrm{seed}}(b) + 1$, then

$$D(F_b) = y_0, \qquad y_0 \leq \mathsf{Rob}(F_b; b, 3/5), \qquad w(F_b) = q_{\mathrm{seed}}. \tag{57}$$

For the upper bound, Bob identifies the coordinate and Alice sends its bit. For the lower bounds, the unit-vector rows give an identity submatrix in the one-output indicator. In a row set of density 2^{-b}, the Boolean vectors span a real space of dimension at least $q_{\mathrm{seed}} - b$. After retaining a $3/(5 \cdot 2^r)$ fraction of columns, the resulting indicator rank exceeds $2^{a_{\mathrm{seed}} - r - 1}$ for $r = 0, 1, 2$, by (55). The fixed-output rank bound proves the three rungs in (57).

5.2 One matrix-and-budget recursion

At level i, we have a matrix $R_i : [m_i] \times [n_i] \to \{0,1\}$ and a budget y_i for the three-colourable case. Initialize R_0 with the canonical zero-based presentation of F_b and the budget above. Feed the level schedule the exact values

$$D_i = y_i + i, \qquad Y_i = y_i, \qquad b_0 = b_+ = b. \tag{58}$$

The schedule also uses the promise-independent moderate exponent

$$h(n) = \max\left\{5, \left(\log_2^{\downarrow}(\log_2^{\downarrow} n)\right)^4\right\}, \tag{59}$$

where the integer floor logarithm is total and is zero at zero. Let H_i be the level overhead from (48). Define

$$R_{i+1} = \mathsf{Succ}(S, R_i; \pi_i), \qquad y_{i+1} = y_i + H_i. \tag{60}$$

The next matrix and its parameters depend on the current matrix and budget. This single sequence is defined before either source condition is used.

Theorem 5.1 (Gap induction)**.** *For every $0 \le i \le k$, once the uniform cutoff conditions hold,*

$$S \textit{ is properly three-colourable} \Longrightarrow D(R_i) \le y_i, \tag{61}$$

$$S \textit{ is in the negative source case} \Longrightarrow \begin{cases} D(R_i) = y_i + i, \\ y_i + i \le \mathsf{Rob}(R_i; b, 3/5). \end{cases} \tag{62}$$

Proof. At level zero, both implications follow from (57). For the successor, the positive induction hypothesis and (49) give $D(R_{i+1}) \le y_i + H_i = y_{i+1}$. On the negative branch, (58) turns the induction hypothesis into the exact seed assumptions needed at level i. Equations (50)–(51) give

$$D(R_{i+1}) = (y_i + i) + H_i + 1 = y_{i+1} + i + 1$$

and regenerate the robust property at depth b. The same matrices and budgets occur in both arguments. ☐

5.3 Exact row and column recurrences

At one level, let

$$\begin{aligned} x_1 &= q_1 m_i, & y_1 &= n_i^{q_1+s+1}, \\ x_2 &= q_2 y_1, & y_2 &= N_2, \\ x_3 &= s y_2, & y_3 &= x_2^s, \end{aligned}$$

and abbreviate

$$\begin{aligned} T &= x_3, & V &= 4s\,v(G) m_i, & B_0 &= 4s(s-3)m_i, & & \\ X &= T + V + B_0, & S_0 &= y_3, & D' &= x_2, & M_0 &= n_i^{s+1} S_0. \end{aligned}$$

Collecting the labelled multiplicities from (26)– (27) gives the exact identities

$$|\mathsf{Rows}(A_i)| = 2^{22} XTVB_0, \qquad |\mathsf{Cols}(A_i)| = 184 M_0 S_0 D'. \tag{63}$$

Since $N_i = A_i^{\mathsf{T}}$ and the two subsequent transformations are interlaces and transpositions, the successor satisfies

$$m_{i+1} = \left(q_{\mathrm{al},i}|\mathsf{Cols}(A_i)|\right)^{Q_{\mathrm{br},i}}, \tag{64}$$
$$n_{i+1} = Q_{\mathrm{br},i}|\mathsf{Rows}(A_i)|^{q_{\mathrm{al},i}}. \tag{65}$$

We use these identities to bound the dimensions of R_{i+1}.

Suppose a number $z \geq 1$ bounds the ceiling logarithm of each of $X, T, V, B_0, M_0, S_0, D'$ and also satisfies $q_{\mathrm{al}}, Q_{\mathrm{br}} \leq z$. Then

$$|\mathsf{Rows}(A)| \leq 2^{22+4z}, \qquad |\mathsf{Cols}(A)| \leq 2^{8+3z}, \tag{66}$$

and (64)– (65) give

$$w(R_{i+1}) \leq A(z) := \max\{1, (4z+8)z, z + (4z+22)z\} = 4z^2 + 23z \leq 27z^2. \tag{67}$$

Choose one branch-uniform initial bound z_0 for the seed, every ingredient, and each interlace width through the fixed k levels, and define $z_{i+1} = A(z_i)$. Induction using the exact carrier equations proves

$$w(R_{i,j}) \leq z_i \quad \text{for every source-list index } j \text{ and every } i \leq k. \tag{68}$$

5.4 One cutoff before the formula

Appendix C gives the parameter choices and the inequalities they satisfy. For fixed k, all parameter widths, ingredient logarithms, and the scalar sequence in (67) are bounded uniformly over the source list by

$$Z_k(n) = 2^{A_k(1+\ell(n))^{d_k}}, \qquad \ell(n) = \log_2^{\downarrow}(\log_2^{\downarrow} n), \tag{69}$$

for constants A_k, d_k. A polynomial in $\log\log n$ is eventually smaller than $\log n$, so for every fixed positive A, d, C there is a cutoff beyond which

$$C\, 2^{A(1+\ell(n))^d} \leq n. \tag{70}$$

Applying this with $C = 2k$ and enlarging the cutoff to cover the source and all structural and numerical conditions yields one $U_k \geq \max\{k, n_{\mathrm{src}}\}$, independent of the formula and branch, for which

$$z_k \leq \left\lfloor \frac{n}{2k} \right\rfloor = e(n,k). \tag{71}$$

Finally, if $w(M) \leq e(n,k)$, then both sides of M have size at most $2^{e(n,k)}$. Their product is at most $2^{2e(n,k)}$, and

$$2\left\lfloor \frac{n}{2k} \right\rfloor \leq \left\lfloor \frac{n}{k} \right\rfloor \tag{72}$$

because $k > 0$. This proves the side and entry bounds for the same matrix $R_{k,j}$ controlled by the gap induction.

6 Proof of the main theorem

We now prove Theorem 1.1. Fix the two auxiliary results and $k \geq 1$. Choose the common cutoff U_k from Section 5, enlarged so that $e(n,k) \geq 1$ beyond it. This choice depends on k and the two fixed inputs, but not on a formula or a source-list index.

Let Φ be a valid formula with $n(\Phi) \geq U_k$. The source theorem gives one nonempty ordered list $\mathcal{S}(\Phi)$. For every $j \in [t(\Phi)]$, run the single recursion (60) on the corresponding source graph and define

$$M_{\Phi,k,j} = R_{k,j}, \qquad Y_{\Phi,k,j} = y_{k,j}. \tag{73}$$

The cutoff and the exact dimension formulas show that all row and column sets are nonempty and every level uses the stated construction. This defines the entire family before we distinguish the two formula cases.

If Φ is satisfiable, source clause (S4) supplies an index j whose graph is properly three-colourable. The positive half of Theorem 5.1 gives

$$D(M_{\Phi,k,j}) = D(R_{k,j}) \leq y_{k,j} = Y_{\Phi,k,j}.$$

If Φ is unsatisfiable, every source graph satisfies (S5). The second part of the same theorem gives, for every j,

$$D(M_{\Phi,k,j}) = D(R_{k,j}) = y_{k,j} + k = Y_{\Phi,k,j} + k.$$

This is an equality: the explicit selector protocols supply the upper bound at every level, while the lower bounds proved for the successive matrices supply the matching lower bound.

For every j, independently of the formula case, (68) and (71) give

$$w(M_{\Phi,k,j}) = w(R_{k,j}) \leq e(n(\Phi), k).$$

The conversion following (72) gives the row, column, and entry bounds for $R_{k,j}$. The source list is nonempty, as required. This completes the proof.

The construction separates the two tasks of amplification and size control. The protected lower bounds and the common-threshold bridge make the one-bit improvement repeatable. The exact dimension formulas then bound the matrices produced by that repetition. Both arguments use one family fixed from the source list, and together yield the exact gap and matrix-size bounds of Theorem 1.1.

A Lower bounds for protection and scale alignment

This appendix gives the construction-specific lower-bound arguments. All row and column sets below are subsets of the matrices defined in Section 3.

We use the *Localised extension* corollary of the companion construction [2], together with its two-copy and odd-copy lower bounds. The four applications below record the substitution rules, not protocol-independent values of every parameter. Their parameters record the number v of coupled streams, a family width K, a retained mass h, and the number of communication bits already charged. The four substitution rules are fixed in advance:

Call	Purpose	v	(K,h)	charged bits
(i)	a protected stop after Alice speaks	1	$(\kappa, x_{\mathrm{Dom}})$	1
(ii)	a terminal pair with one ordinary and one foreign stream	2	$(\kappa, 49/64)$	1
(iii)	protected donor mass that stops when Bob first speaks	2^{e-1}	$(s\kappa q_2, \rho_1 2^{-(2^e)-1})$	0
(iv)	the exceptional terminal half-block	$2^{e_{\mathrm{exc}}-1}$	$(2sq_2, \rho_0 2^{-(2^{e_{\mathrm{exc}}})-1})$	0

Here κ is a remaining coordinate capacity at the selected protocol node v, whose path has length d_v. In (i)–(ii), $\kappa = 2^{r_2-d_v}$; in (iii), $\kappa = 2^{r_{\mathrm{al}}-d_v}$. Thus κ depends on where the protocol stops. For (i)–(ii), the localised extension corollary applies when $\kappa \geq t_2$. Capacities $2 \leq \kappa < t_2$ use the inherited two-copy and odd-copy bounds. At $\kappa = 1$, the stopping count instead gives either a hard reference witness whose path never split, or two ordinary coordinates to which the two-copy bound applies.

To define the fraction in (i), let N_2 be the number of indexed members of the Stage–2 family and put

$$T_0^{\mathrm{aux}} = \lceil 999N_2/1000 \rceil, \qquad T_3^{\mathrm{aux}} = \left\lceil 2^{1-b_T} N_2 \right\rceil, \qquad x_{\mathrm{Dom}} = \frac{T_0^{\mathrm{aux}} - (s-1)T_3^{\mathrm{aux}}}{N_2}. \tag{74}$$

This lower-bounds the dominant-block fraction after reserving up to T_3^{aux} rows in each of the other $s-1$ blocks. In (iii), the selected dyadic donor class has width 2^e with $e \geq 1$; in (iv), the canonical exceptional class has width $2^{e_{\mathrm{exc}}}$ with $e_{\mathrm{exc}} \geq 2$. Thus the half-blocks in the table have the displayed cardinalities. The fixed projection fractions are $\rho_0 = 99/100$ and $\rho_1 = \rho_0/2 = 99/200$. A selected donor projection retains at least fraction ρ_0; after Bob's query, one child retains at least ρ_1. In each application, the row and column maps take values in the retained matrix, (B1), (B2), or (B3) gives the required balanced-family count, and the parameter inequalities give the retained mass. The corresponding lower bound for the input matrix accounts for the communication bits.

A.1 Protected Stage 4

Fix a retained restriction of N keeping a $(1-\eta_R)$ fraction of rows and a $(1-\eta_C)$ fraction of columns, and suppose that a protocol for it has cost below C. Because the rows and columns of $A = N^{\mathsf{T}}$ were duplicated into labelled groups, we can bound the deletion loss separately in each group. The inequalities in (114) imply:

(P1) all but a $1/1000$ fraction of template rows keep enough labels;

(P2) all but a $\delta_V/16$ fraction of vertex-source rows keep enough labels;

(P3) all but a $1/(16(s-3))$ fraction of blocker-source rows keep enough labels;

(P4) every selector and diagonal column class required by the proof has a surviving representative.

These are integer counts obtained by clearing positive denominators.

The ordinary balanced-family count then leaves a set of good Stage–2 coordinates on which the singleton, pair, and reference patterns required by the proof all occur. If too many coordinates were bad, the missing family members would exceed the total deletion allowance. Projection covers make the exceptional set finite and give the bound $e_{\max}$ used by the schedule.

Follow the supposed protocol while keeping surviving template rows. At a reached node, remember the compatible row and column sets, represented good coordinates, their selector and diagonal representatives, the partial source colouring fixed so far, and a debt in $\{0, 1, 2\}$ recording which profile rung is being used. At an Alice node, her query partitions the row groups; counting the surviving labels selects a nonempty child with the required mass. At a Bob node, his query partitions columns; a selector or diagonal representative either survives in a child or spends one unit of profile debt.

If this routing could not synchronize a represented coordinate, its answers would define a three-colouring of G with fewer than $\gamma e(G)$ monochromatic edge occurrences. Vertex rows enforce consistency at the two incidence positions, blocker rows keep the unused colours distinct, and

$$\frac{64s(2^a-2)}{\gamma/24} \le q_2 \tag{75}$$

ensures that the discarded coordinates account for less than a γ fraction of edge occurrences. This contradicts the negative source condition.

Stop at the first Bob node or when all selector bits of a represented coordinate have been exposed. The stopping routes are prefix-free, their row sets are disjoint, and Kraft's inequality bounds their total binary capacity by one. Counting rows over these stopping paths therefore gives a stop with enough template and source-owned mass. If all selector bits have been exposed, the surviving selector and diagonal columns embed the appropriate incoming seed rung. If Bob speaks first, the protected one-stream extension lemma embeds a hard restriction in one of his real children. Either suffix requires at least $C-d$ further bits after a route of length d, contradicting the alleged protocol. This proves (34).

For the reference rungs, retain only the base-labelled raw columns that form the rows of N_{ref}. At rung $r \in \{0,1,2\}$, the exact template density and the ceiling inequality in (116) leave at least $9s/(16 \cdot 2^r)$ Stage–3 blocks. The reference map sends these rows and all retained columns into the restriction of N_{ref}. The fixed reference extension theorem embeds the appropriate seed rung and gives cost at least $C-1-r$. This proves (36).

A.2 Stopping paths for scale alignment

Consider a restriction of $U = \mathrm{Int}_{q_{\mathrm{al}}}(N)$ with row density $1-\delta_U$ and column density θ. After removing coordinates with excessive physical or reference-row loss and those excluded by the three projection covers, obtain an exceptional set E_0 of size at most $e_{\max}$. Each coordinate outside E_0 has the initial reference floor and every column projection needed below.

Suppose that a protocol Q has cost at most $C_W - 1$. Follow its Alice questions until Bob first speaks or until $r_{\mathrm{al}} = \log_2 q_{\mathrm{al}}$ Alice answers have been seen. The paths stopping when Bob first speaks, together with the final length-r_{al} path labels, form a prefix-free set $\mathcal{T}$. If d_v is the route length and $u_v = 2^{r_{\mathrm{al}}-d_v}$ its residual dyadic capacity, then

$$R_U = \bigsqcup_{v\in\mathcal{T}} \mathcal{R}_v, \qquad \sum_{v\in\mathcal{T}} 2^{-d_v} \le 1, \qquad \mathrm{cost}(Q_v) \le C_W - 1 - d_v. \tag{76}$$

For each coordinate outside E_0, follow a reference-row witness through the Alice queries. If its threshold at remaining depth k is $A_k = 2^k A_0$, at least one child contains A_{k-1} reference rows; keep both children when both qualify and otherwise keep the unique qualifying child. Stop these paths at $\mathcal{T}$. A reference-row witness, which we also call a *certificate*, records its coordinate, stopping path, reference-row subset, and whether it split. We count the protected rows from that coordinate separately: they may reach several stopping paths.

At a node where Bob first speaks, small residual capacity is contradicted by one of the three reference bounds. Large residual capacity provides a dyadic class of coordinates contributing rows, called *donors*. Product estimate (B1), the row and column maps, and the multi-stream extension lemma give a hard restriction in one child. Thus only a bounded amount of protected mass associated with clean certificates can stop early.

A.3 Counting rows at the final paths

The last step compares the paths followed by the reference witnesses with the distribution of all protected rows. These are different quantities: a witness follows a specified path, while rows from its coordinate may reach several paths. Put $\vartheta = 9/16$. Let J_{clean} be the clean reference witnesses ending at final path labels, let $c = |J_{\mathrm{clean}}|$, let n_T be the total number of final length-r_{al} path labels, and put $K_B = \sum_{v \text{ first Bob}} u_v$. Certificate conservation and Kraft's inequality give

$$c \geq q_{\mathrm{al}} - \frac{e_{\max}}{1-\vartheta}, \qquad K_B + n_T \leq q_{\mathrm{al}}. \tag{77}$$

Normalize protected row mass by $m = |\mathsf{Rows}(N)|$. If $P_v(J)$ is the mass from the coordinates of witnesses in J reaching a path v where Bob first speaks, the contrapositive of the heavy-node extension gives

$$P_B := \sum_{v \text{ first Bob}} P_v(J_{\mathrm{clean}}) < 128\mu B_{\mathrm{pair}} L_q s K_B \leq \frac{128\mu B_{\mathrm{pair}} L_q s e_{\max}}{1-\vartheta}. \tag{78}$$

Call a witness *Bob-heavy* if the number of protected rows from its coordinate that stop when Bob first speaks, divided by $m = |\mathsf{Rows}(N)|$, exceeds $\eta_R/4$. Markov counting and (78) give

$$e_B \leq \frac{512\mu B_{\mathrm{pair}} L_q s e_{\max}}{(1-\vartheta)\eta_R} \leq E_B. \tag{79}$$

The remaining certificates are the good donors J_{good}.

The owner $o(j)$ of a good donor is the final path label at which its clean witness stops. The uniqueness of final reference witnesses makes o injective. The exceptional final path labels are those outside $o(J_{\mathrm{good}})$; if their count is e_T, then

$$e_T \leq \left\lceil \frac{e_{\max}}{1-\vartheta} \right\rceil + E_B = E_T. \tag{80}$$

The schedule ensures $E_T \leq q_{\mathrm{al}}/8$, and therefore $|J_{\mathrm{good}}| \geq 7q_{\mathrm{al}}/8 > q_{\mathrm{al}}/2$.

For a donor j and final path label T, let $m_{j,T}$ be the normalized physical protected mass from j that reaches T. Mass missing from the owner is partitioned into initially deleted rows, rows lost from the unique clean reference-witness path, rows stopped when Bob first speaks, and rows reaching other final path labels. The first three charges are at most $\eta_R/16$, strictly below $\eta_R/16$, and at most $\eta_R/4$, respectively. If the last charge were at most $\eta_R/2$, the owner would retain more than $1-\eta_R$ of its physical copy of N and a $1-\eta_C$ column projection. The protected theorem would give residual cost at least C, while the remaining protocol at a final path has cost at most $C-1$. Hence

$$\sum_{T \neq o(j)} m_{j,T} > \frac{\eta_R}{2} \qquad (j \in J_{\mathrm{good}}). \tag{81}$$

It remains to control transfers to other good owners. The ordinary baseline occupies the fraction $g = 91/92$ of one physical copy. Let ε_i be its deficit at $o(i)$ and define

$$\lambda_0 = q_2 2^{1-b_1}, \qquad \delta_i = \lambda_0 + 5\varepsilon_i + \frac{2}{q_2}. \tag{82}$$

Disjoint owner baselines give

$$\sum_{i \in J_{\text{good}}} \varepsilon_i \le \frac{\eta_R}{8g} < \frac{1}{5}. \tag{83}$$

For distinct good donors j, i, if $m_{j,o(i)} \ge 2\mu\delta_i$, product clause (B2) and the two-stream extension construct a restriction at $o(i)$ of cost at least C. This contradicts its $C-1$ suffix budget, so

$$m_{j,o(i)} < 2\mu\delta_i. \tag{84}$$

The three inequalities in (118) and (83) now yield

$$\sum_{\substack{i \in J_{\text{good}} \\ i \ne j}} m_{j,o(i)} < 2\mu \sum_i \delta_i \le \frac{3\eta_R}{16} < \frac{\eta_R}{4}. \tag{85}$$

Combining this with (81), every good donor sends more than $\eta_R/4$ of its mass to exceptional final paths.

If there are no exceptional final paths, this is impossible. Otherwise, two applications of the pigeonhole principle choose an exceptional path label T_*, one of the two protected row-group types, and a donor set J_* satisfying

$$|J_{\text{good}}| \le 2e_T|J_*|, \qquad \delta_T \le \frac{\eta_R}{8\mu e_T} \le \text{density}_{T_*}(j) \quad (j \in J_*). \tag{86}$$

The inequalities

$$4E_T u_T B_{\text{pair}} < q_{\text{al}}, \qquad u_T\delta_T \ge 8s \tag{87}$$

leave u_T donors with the required density after projection-cover loss. Choose a half-block of size $u_T/2$. Product clause (B3) and the exceptional-terminal extension construct a restriction of the remaining protocol at T_* with cost at least C, contradicting its cost at most $C-1$. This completes the proof of the scale-aligned lower bound.

B Defect induction for the generic bridge

We prove Theorem 4.2. Expanding the low-profile part of (43) and applying the inherited finite two-copy ladder gives

$$C_W \le \Lambda_W(2; 2\theta, (3/5)^2). \tag{88}$$

Because $4\theta < 1$, the inherited three-copy row step gives

$$C_W + 1 \le \mathcal{D}_W(3; 4\theta, (3/5)^2). \tag{89}$$

These two estimates, together with the near-full estimate, are the communication lower bounds used in the bridge.

Write $m = |\mathsf{Rows}(W)|$, $n = |\mathsf{Cols}(W)|$, and

$$\lambda = -\ln(1-\delta), \qquad A_a = \lceil m2^a\theta \rceil. \tag{90}$$

For a nonempty $K \subseteq \mathsf{Cols}(W)^p$, define its logarithmic defect

$$\mathrm{Def}_p(K) = \ln \frac{n^p}{|K|}. \tag{91}$$

Projecting K to some p' of its p coordinates can be done with defect at most $(p'/p)\,\mathrm{Def}_p(K)$. If the coordinates are partitioned into two sets, the sum of the two projection defects is at most $\mathrm{Def}_p(K)$. Both facts follow from finite product counting.

For $p \geq 2$ let

$$B(p) = \left(1 + \frac{6}{5p}\right) \ln 2. \tag{92}$$

The elementary inequalities needed in the induction are

$$\begin{aligned} &B(p) \geq \ln 2, && B(p') \geq B(p) \quad (2 \leq p' \leq p), \\ &B(\lceil p/2 \rceil) \geq \frac{\lceil p/2 \rceil}{p}(B(p) + \ln 2), && \frac{3}{p}B(p) \leq \frac{7}{5}\ln 2. \end{aligned} \tag{93}$$

For $a \geq 0$ and $p > 2^{a-1}$, with $2^{-1} = 1/2$, put

$$c_a(p) = p - 2^{a-1}. \tag{94}$$

Lemma B.1 (Defect induction)**.** *Assume*

$$3\lambda + \frac{7}{5}\ln 2 \leq -2\ln(3/5). \tag{95}$$

Let $a \geq 0$, $p > 2^{a-1}$, and $2^a\theta \leq 1$. Every restriction of $\mathrm{Int}_p(W)$ that keeps at least A_a rows in each active copy and has nonempty column set K satisfying

$$\mathrm{Def}_p(K) \leq c_a(p)\lambda \tag{96}$$

has communication complexity at least $C_W + a$.

The proof uses the following one-bit-down auxiliary statement: for $a \geq 2$, the relaxed condition

$$\mathrm{Def}_p(K) \leq c_a(p)\lambda + B(p) \tag{97}$$

implies cost at least $C_W + a - 1$.

At $a = 2$, project to three copies. Its defect is at most $3\lambda + (3/p)B(p)$, hence at most $-2\ln(3/5)$ by (93) and (95). The column density is therefore at least $(3/5)^2$, and (89) gives cost at least $C_W + 1$.

For the auxiliary induction step, consider the first protocol bit. If Alice speaks, assign each active copy to a child retaining at least half its rows; since $A_{a-1} \leq \lceil A_a/2 \rceil$, that child has the next row floor. If p_i and L_i are the number of copies and projection defect assigned to child i, then $p_0 + p_1 = p$ and $L_0 + L_1 \leq \mathrm{Def}_p(K)$. The inequality

$$(p_0 - 2^{a-2})_+ + (p_1 - 2^{a-2})_+ \geq p - 2^{a-1}$$

forces a child satisfying the auxiliary hypothesis at level $a - 1$. If Bob speaks, keep a child containing at least half of K, increasing defect by at most $\ln 2$, and project to $p' = \lceil p/2 \rceil$. Direct substitution and (93) give

$$\frac{p'}{p}c_a(p) \leq c_{a-1}(p'), \qquad \frac{p'}{p}(B(p) + \ln 2) \leq B(p').$$

The auxiliary induction hypothesis again applies to a child of the protocol. Adding the first bit proves the auxiliary statement.

For the main lemma, the case $a = 0$ projects to one copy with defect strictly below λ, so the column density exceeds $1 - \delta$ and the near-full common-threshold estimate gives C_W. At $a = 1$, an Alice first bit reduces to $a = 0$ by the same split. A Bob first bit keeps half the columns and projects to two copies; its defect is at most $2\lambda + \ln 2 \leq -2\ln(3/5)$, so (88) applies. For $a \geq 2$, an Alice first bit yields the main induction hypothesis in one child; a Bob first bit yields the auxiliary statement at the same level. In both cases the parent costs at least $C_W + a$. This proves Theorem B.1.

B.1 Entering the induction at the requested depth

Put $Q = \mathsf{Qwidth}(b, \delta)$, $r_Q = \log_2 Q$, and $L = b \ln 2$. Since $\lambda \geq \delta$ and $80b \leq Q\delta$,

$$Q \geq \frac{80L}{\lambda}, \qquad Q \geq 80. \tag{98}$$

The bound $\delta \leq 1/64$ implies the fixed margin (95).

Let d be the least positive integer for which $(1-\delta)^d \leq 2^{-b}$. Its minimality says

$$(d-1)\lambda < L \leq d\lambda. \tag{99}$$

For $t = 0, 1, 2$, set

$$a_t = r_Q - t, \qquad p_t = 2^{a_t - 1} + d. \tag{100}$$

Then $p_t > 2^{a_t-1}$ and $p_t \leq Q$. The following count includes the necessary ceiling terms:

$$(p_t - 1)m + (Q - p_t + 1)(A_{a_t} - 1) < Qm\frac{3/5}{2^t}. \tag{101}$$

After division by Qm, the left side is strictly below

$$2^{-t-1} + \frac{L}{Q\lambda} + \frac{Q\theta}{2^t} < \frac{3/5}{2^t},$$

using $L/(Q\lambda) \leq 1/80$ and $Q\theta < 1/20$.

Take any retained member in the t-th rung for $R^+ = \mathrm{Int}_Q(W)^{\mathsf{T}}$ and transpose it. Its row count and (101) leave at least p_t outer copies with A_{a_t} rows. Project its retained columns to those copies. The column density is still at least 2^{-b}, so its defect is at most L. Moreover,

$$2^{a_t}\theta = Q\theta/2^t < 1, \qquad L \leq d\lambda = c_{a_t}(p_t)\lambda.$$

The defect induction gives cost at least $C_W + r_Q - t$. These three values are exactly the rungs in (46). The full-matrix $t = 0$ case gives the lower bound on $D(R^+)$, and the selector protocol gives the matching upper bound as well as (47).

C Parameter choices and quantitative estimates

This appendix gives one generous choice of parameters satisfying all inequalities used in the proof. The parameters depend on the source graph and the current matrix, not on a colouring or on a claim that the graph is far from three-colourable.

C.1 Choosing the parameters

Besides the current matrix, use $I = (D, Y, b_0, b_+)$ and a natural exponent h. Define

$$\begin{aligned}
\sigma &= \max\{6, h+2\}, & s &= 2^{\sigma}, \\
r_2 &= \lceil \log_2(12se(G)) \rceil, & q_2 &= 2^{r_2}, \\
\ell_1 &= \lceil \log_2(r_2 + 18) \rceil, & u_1 &= \ell_1 + 2, \\
a &= h + 2 + 16u_1^2, & q_1 &= 2^a - 2, \\
b_1 &= 2^{u_1}, & b_2 &= \lceil \log_2(32sq_1m_R + 2) \rceil + 6.
\end{aligned} \tag{102}$$

Put

$$\mu = \frac{1}{184}, \qquad \zeta = \frac{\gamma}{24}, \qquad \delta_V = \frac{\gamma e(G)}{2\Delta v(G)}, \tag{103}$$

and then

$$\eta_C = \frac{1}{4}\min\left\{\frac{2^{-21}}{1000s}, \frac{2^{-22}\delta_V}{16}, \frac{2^{-22}}{16(s-3)}\right\}, \tag{104}$$

$$\eta_R = \frac{1}{4}\min\left\{\frac{\mu}{100}, \frac{\mu\zeta}{32s}, \eta_C\right\}, \qquad n_\eta = \max\{2, \lceil 1/\eta_R\rceil\}. \tag{105}$$

For the reference and alignment widths, set

$$\begin{aligned}
U_{\mathrm{ref}} &= \max\{16, D + Y + b_0 + b_+ + r_2 + \sigma + a + n_\eta + 64\}, \\
L_{\mathrm{ref}} &= \lceil \log_2 U_{\mathrm{ref}} \rceil, \\
k_{\mathrm{ref}} &= 2^{20} + 4096(L_{\mathrm{ref}} + 1)^2, \\
r_{\mathrm{al}} &= 64(k_{\mathrm{ref}} + L_{\mathrm{ref}} + \sigma + 1), \\
q_{\mathrm{al}} &= 2^{r_{\mathrm{al}}}, & b_T &= 2r_{\mathrm{al}} + L_{\mathrm{ref}} + 32, \\
\delta_U &= \frac{\eta_R}{16 q_{\mathrm{al}}}, & Q_{\mathrm{br}} &= \mathsf{Qwidth}(b_+, \delta_U), \\
\theta &= \frac{1}{160 Q_{\mathrm{br}}}.
\end{aligned} \tag{106}$$

For $0 \le \rho < 1$ and $\theta > 0$, define

$$\begin{aligned}
B(\rho,\theta) &= \min\{B \in \mathbb{N} : B \ge 1,\ \rho^B < \theta\}, \\
(B_{\mathrm{ref}}, B_{\mathrm{sing}}, B_{\mathrm{pair}}) &= \big(B(18/25,\theta), B(1-\eta_C,\theta), B(99/100,\theta)\big).
\end{aligned} \tag{107}$$

The minimum exists because ρ^B tends to zero. These exponents bound exceptional coordinate sets as follows. If a set of tuples in a finite product of nonempty sets has density at least θ, there is one coordinate set E of size at most $j(B(\rho,\theta)-1)$ that meets the coordinate set of every j-coordinate projection of density below ρ. Indeed, put $B = B(\rho,\theta)$ and take a maximal disjoint collection of such projections: B of them would force the original density below $\rho^B < \theta$. We use this observation for the small-reference, single-coordinate, and paired-coordinate projections, respectively. Define

$$\begin{aligned}
t_{\mathrm{ref}} &= 2^{k_{\mathrm{ref}}}, \\
e_{\max} &= (B_{\mathrm{ref}} - 1)(t_{\mathrm{ref}}/2 + \log_2 t_{\mathrm{ref}} - 2) + (B_{\mathrm{sing}} - 1) + 2(B_{\mathrm{pair}} - 1), \\
L_q &= \lceil \log_2(2q_{\mathrm{al}})\rceil + 1, \\
E_B &= \left\lceil \frac{512\mu B_{\mathrm{pair}} L_q s e_{\max}}{(1-9/16)\eta_R} \right\rceil, \\
E_T &= \left\lceil \frac{e_{\max}}{1-9/16} \right\rceil + E_B, \\
\delta_T &= \frac{\eta_R}{8\mu \max\{1, E_T\}}, \\
u_T^* &= \max\{1, \lceil 8s/\delta_T \rceil\}, & u_T &= 2^{\lceil \log_2 u_T^* \rceil}.
\end{aligned} \tag{108}$$

Finally,

$$
\begin{aligned}
L_h &= \max\{b_2, b_T\} + \max\{q_{\mathrm{al}}, u_T\} + 4, \\
H_{\max} &= L_h + 8q_1 + r_{\mathrm{al}} + r_2 + (D+a) + 10, \\
L_2 &= \max\left\{2, \lceil \log_2(H_{\max}+2) \rceil\right\}, \\
k_2 &= a + h + 4096L_2^2 + 8, \qquad\qquad t_2 = 2^{k_2}.
\end{aligned} \tag{109}
$$

These formulas determine the six parameters in (18). If the source and seed carriers are positive, $D \geq 3$, $b_+ \geq 1$, $h \geq 5$, and

$$k_2 + 2 \leq r_2, \tag{110}$$

then the structural conditions in (19) hold. In particular, $4t_2 = 2^{k_2+2} \leq 2^{r_2} = q_2$, the guarded ranks are below their displayed width, and all three interlace widths are positive powers of two.

C.2 Five comparisons uniform over the input formula

The lower-bound argument needs five further comparisons in addition to the structural conditions. All five hold beyond one cutoff uniform over the input formula:

$$m_{\mathrm{HS}} \leq h, \qquad \frac{64sq_1}{\zeta} \leq q_2, \qquad u_T \leq q_{\mathrm{al}}, \qquad a + 2 + \sigma \leq b_0, \qquad 100(V_0 + B_0) \leq T_0. \tag{111}$$

Here m_{HS} is the least integer $r \geq 5$ such that the following hard-seed conclusion holds for every integer $m \geq r$, every finite nonempty Boolean matrix M, and every real $b \geq 1$ satisfying $m \leq b$ and $3 \leq D(M) \leq \mathsf{Rob}(M; b, 3/5)$:

$$\mathcal{D}_M\left(\frac{9}{16}2^m;\ 2^{m-b},\ 2^{-2^{(49/100)\sqrt{m}}}\right) \geq D(M) + m. \tag{112}$$

Such a uniform threshold exists by the hard-seed lemma of [2], with copy-depth parameter 5 and margin 1/10. The threshold is an exponent, not the width $2^{m_{\mathrm{HS}}}$. The quantities T_0, V_0, B_0 are the numbers of template, vertex, and blocker rows. The balanced count on t_2 coordinates makes every $(q_1 m_R)^{t_2}$ pattern occur, so $N_2 \geq (q_1 m_R)^{t_2}$; this is the lower bound used to show that template rows outnumber the other two groups by the required factor.

Together with the parameter definitions, these five inequalities imply the remaining estimates, grouped below by their use in the proof.

Raw upper protocol. The choice of s makes it a positive power of two and gives

$$s = 2^{\log_2 s}, \qquad s \leq 2^a. \tag{113}$$

Protected losses.

$$
\begin{aligned}
&8q_1 \leq t_2, && \frac{64sq_1}{\zeta} \leq q_2, \\
&\eta_R < \frac{(1/4)\mu\zeta}{8s}, && \eta_C \leq \frac{1}{2}\min\left\{\frac{2^{-21}}{1000s}, \frac{2^{-22}\delta_V}{16}, \frac{2^{-22}}{16(s-3)}\right\}.
\end{aligned} \tag{114}
$$

Reference blocks. The parameter choices give

$$1 \le k_{\mathrm{ref}}, \quad m_{\mathrm{HS}} + 2 \le \log_2 s < b_T, \quad k_{\mathrm{ref}} + 2 \le \log_2 q_{\mathrm{al}}, \quad q_{\mathrm{al}} 2^{-b_T} \le \frac{4}{25025}, \quad a + 2 + \log_2 s \le b_0. \tag{115}$$

Let $w_j = s/2^j$, $T_j = 9w_j/16$, $d_j = w_j 2^{-b_T}$, and

$$\beta_j = \frac{(3/5)/2^j + \tau_{\mathrm{act}} - 1}{\tau_{\mathrm{act}}}, \qquad \tau_{\mathrm{act}} = (1 - 2^{-20}) \frac{T}{T + V + B_0} + 2^{-22}.$$

For $j = 0, 1, 2$, we have the ceiling inequality

$$T_j \le \left\lceil \frac{s(\beta_j - d_j)}{1 - d_j} \right\rceil. \tag{116}$$

The ceiling covers the whole quotient. This is the extremal-bin count used for the three reference lower bounds.

Bridge and owner transfer. The definitions give

$$\delta_U q_{\mathrm{al}} = \eta_R/16, \qquad 80 b_+ \le Q_{\mathrm{br}} \delta_U, \qquad Q_{\mathrm{br}} \theta = 1/160 < 1/20, \qquad \delta_U \le 1/64. \tag{117}$$

With $g = 91/92$ and $\lambda_0 = q_2 2^{1-b_1}$, the three owner-transfer comparisons are

$$2\mu q_{\mathrm{al}} \lambda_0 \le \frac{\eta_R}{16}, \qquad \mu \le \frac{g}{20}, \qquad \frac{64 \mu q_{\mathrm{al}}}{\eta_R} \le q_2. \tag{118}$$

They follow from the cleared natural comparisons $\eta_R^{-1} \le q_{\mathrm{al}}$, $4q_{\mathrm{al}}^2 \le q_2$, and $q_2^2 \le 2^{b_1}$.

Exceptional terminal. The final counting estimates give

$$8 \max\{1, E_T\} \le q_{\mathrm{al}}, \qquad 4 E_T u_T B_{\mathrm{pair}} < q_{\mathrm{al}}, \qquad 4 t_{\mathrm{ref}} \le q_{\mathrm{al}}, \qquad u_T \delta_T \ge 8s. \tag{119}$$

For example, if $K = k_{\mathrm{ref}} + \sigma$, the schedule bounds $E_T \le 2^{6K+51}$, $u_T \le 2^{8K+58}$, and $B_{\mathrm{pair}} \le 2^{K+15}$, while $15K + 126 < r_{\mathrm{al}}$. This proves the strict middle inequality without decimal rounding. Also $e_{\mathrm{exc}} = \log_2 u_T \ge 2$ is fixed before a protocol is chosen, as required by product estimate (B3).

C.3 Why one uniform cutoff exists

For fixed k, induct over the k-level matrix sequence. Finite sums, products, maxima, and fixed powers of quantities bounded by $2^{A(1+\ell)^d}$ remain within another bound of the same form. Taking a ceiling logarithm makes the bound polynomial in $1 + \ell$. Every exponentiation in the schedule is applied either to such a logarithm or to a fixed polynomial in such logarithms. The balanced-family size bound (13) has the same closure property, and the exact carrier recurrence is controlled by the quadratic arrow (67).

Taking a maximum over the finite source list before using either source promise produces the branch-uniform envelope (69). The hard-seed row follows because $\ell(n)^4$ tends to infinity. The Stage–2 fit, protected source width, and input-depth rows reduce to a fixed polynomial in $1 + \ell(n)$ being at most $\log_2^{\downarrow} n$. The terminal definitions first bound the projection exponents and $e_{\max}$, after which the deliberately larger alignment exponent gives $u_T \le q_{\mathrm{al}}$. The balanced-family lower count gives the required proportion of template rows. These establish the five inequalities in (111); all other estimates above follow from them.

The elementary comparison (70) therefore provides one cutoff that simultaneously proves structural legality, every numerical lower-bound condition, nonempty row and column sets, and the final naming-width bound. It uses no monotonicity assumption about how the source list changes with the formula.

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